

PORETSKOYE, Russian Federation

WMO: 276750

Lat: 55.183N

Lon: 46.333E

Elev: 446

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-10.3	-20.5	1.8	-15.1	-15.7	2.4	-10.1	24.1	19.3	21.2	19.5	4.8	200	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	88.4	67.9	84.6	66.3	81.4	65.3	70.6	84.5	68.8	81.1	67.2	78.2	7.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	96.9	76.1	64.4	92.0	73.8	62.8	87.0	72.6	34.7	84.6	33.2	80.8	31.9	78.5	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.4	16.3	DB	-21.5	91.9	6.0	3.9	-25.8	94.7	-29.3	97.0	-32.7	99.2	-37.1	102.1	(4)
(5)			WB	-21.7	73.9	5.9	2.8	-26.0	75.9	-29.5	77.6	-32.8	79.2	-37.1	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	11.5	15.3	24.7	42.5	58.8	63.8	68.0	65.4	54.4	40.6	29.9	20.3		
	DBStd	21.81	12.74	12.09	8.95	8.87	8.16	7.55	6.49	7.48	7.37	8.43	8.68	11.38		
	HDD50	5104	1193	971	785	260	18	3	0	1	39	308	604	922		
	HDD65	8953	1658	1391	1250	674	223	110	40	86	325	755	1054	1387		
	CDD50	1971	0	0	0	37	292	418	559	477	170	18	0	0		
	CDD65	345	0	0	0	0	32	75	134	98	6	0	0	0		
	CDH74	3350	0	0	0	4	350	732	1273	916	75	0	0	0		
	CDH80	1067	0	0	0	0	66	225	445	323	8	0	0	0		
	WSAvg	8.0	8.9	9.3	8.9	8.6	7.9	7.1	6.3	6.3	6.7	8.3	8.8	9.4		
	PrecAvg	20.3	1.0	0.8	0.8	1.4	1.5	2.6	2.8	2.4	2.0	1.9	1.5	1.1		
(15) Precipitation	PrecMax	27.9	2.6	2.4	2.7	4.0	4.6	5.3	5.9	8.2	5.2	5.9	2.8	2.1		
	PrecMin	14.0	0.2	0.0	0.1	0.2	0.0	0.9	0.0	0.0	0.2	0.0	0.3	0.1		
	PrecStd	3.8	0.6	0.5	0.5	0.9	1.1	1.1	1.5	1.5	1.2	1.1	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	35.8	46.7	73.8	85.7	89.9	96.5	96.4	81.0	65.1	52.8	39.6		
		MCWB	32.9	33.3	42.2	56.1	62.2	67.8	70.7	73.0	60.6	55.3	48.4	38.0		
	2%	DB	33.3	34.2	40.8	68.7	81.6	85.8	89.1	87.7	75.6	59.7	48.9	36.6		
		MCWB	31.9	32.2	36.2	53.1	61.3	66.9	69.1	67.4	59.5	52.3	46.1	35.1		
	5%	DB	31.8	33.0	37.8	63.2	78.2	82.1	84.7	82.7	71.2	55.7	44.7	34.7		
		MCWB	30.8	31.3	34.2	50.4	60.4	66.1	67.6	66.1	57.9	50.4	42.8	33.2		
	10%	DB	29.4	30.9	35.7	58.0	74.0	77.9	80.9	78.7	66.7	52.4	41.2	33.2		
		MCWB	28.3	29.3	33.1	47.7	58.5	63.5	66.4	64.9	56.4	48.2	39.5	31.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	34.0	42.7	58.2	66.3	71.1	73.6	73.3	63.7	56.4	49.4	38.0		
		MCDB	34.2	35.3	44.7	69.1	78.0	83.7	88.8	95.5	73.9	62.6	52.2	39.1		
	2%	WB	32.3	32.6	36.7	55.0	63.7	69.2	71.0	70.0	61.3	53.4	46.1	35.5		
		MCDB	33.3	33.9	39.4	65.2	77.0	82.1	85.8	83.1	71.4	58.5	48.8	36.3		
	5%	WB	30.7	31.5	34.7	51.6	62.0	67.6	69.2	67.9	59.5	51.0	42.9	33.4		
		MCDB	31.8	32.9	36.9	62.3	75.2	79.1	81.7	79.3	67.9	54.7	44.6	34.3		
	10%	WB	28.3	29.3	33.3	48.4	59.9	65.4	67.5	66.0	57.6	48.6	39.7	32.1		
		MCDB	29.6	31.0	35.6	57.3	71.9	75.2	78.7	75.3	65.0	52.0	41.0	33.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.4	12.4	16.1	19.6	18.4	19.0	18.8	16.0	10.6	7.5	8.2		
		MCDBR	7.2	8.6	12.3	24.4	25.1	23.7	24.8	24.6	24.7	15.7	9.4	6.5		
		MCWBR	6.8	7.7	9.1	12.4	10.9	9.9	10.0	9.5	11.2	10.1	8.2	6.4		
		MCDBR	7.2	8.0	11.1	22.4	22.4	20.9	21.7	21.4	20.3	13.5	8.6	6.6		
	5% WB	MCWBR	6.9	7.6	8.9	12.0	11.1	9.7	10.2	9.6	10.3	10.0	7.9	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.322	0.358	0.414	0.388	0.382	0.403	0.427	0.376	0.352	0.318	0.289		
	taud		2.083	2.047	2.025	2.117	2.319	2.368	2.315	2.232	2.359	2.330	2.223	2.072		
	Ebn at Noon		193	229	250	254	271	274	265	248	247	222	183	160		
	Edh at Noon		21	31	41	43	37	36	37	38	30	24	18	17		
(44) All-Sky Solar Radiation	RadAvg		229	528	1019	1312	1743	1838	1831	1444	897	424	170	126		
	RadStd		33	65	99	132	119	165	169	135	124	69	32	33		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	-301	N/A	N/A	

Nomenclature: See separate page